



Finding the percentage change

Task A: Finding the percentage change

1) Calculate the percentage change from the old to the new values.

Old	New
55	44
60	69
45	72
20	11
150	112.5
82	83.64
60	58.2
18	18.9
54	70.2

Shade in the answers in the grid to reveal an image.

Increases are identified by an I and decreases by a D.

20% D	25% D	2% I	2% D	20% I	5% D	22% D	15% I	3% I
45% D	60% D	60% I	35% I	25% I	45% I	20% D	55% I	60% D
5% I	3% D	30% I	15% D	42% D	45% D	12% I	25% I	2% D
45% I	42% D	60% D	55% D	60% I	55% D	35% I	45% I	22% D
12% I	3% I	55% I	2% I	12% I	25% I	3% D	2% I	15% I
20% I	35% I	25% D	22% D	5% D	3% I	30% I	2% D	25% D
5% D	30% I	55% D	15% D	55% I	20% I	60% I	5% I	3% D

2) Below are some items that were bought and sold by a market stall holder. Rank the items based on the percentage change.



Item	Purchase price	Selling price	Percentage change
Mobile phone	£120	£105.60	12% decrease
Sunglasses	£12	£13.20	10% increase
Tablet	£160	£172.80	8% increase
Shoes	£12	£11.40	5% decrease
Cap	£8	£9.20	15% increase

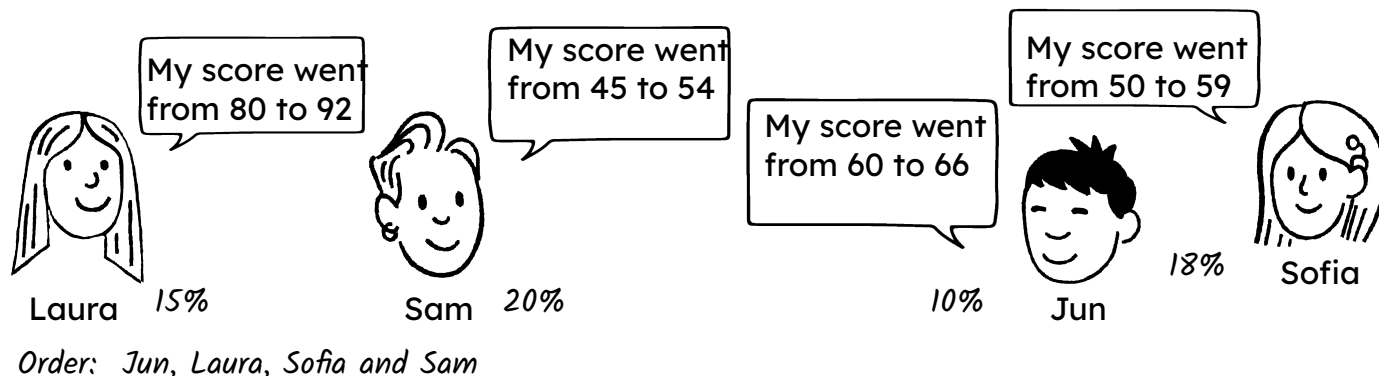
Order: mobile phone, shoes, tablet, sunglasses and cap (or reverse).



Task B: Using multipliers for percentage change

1) Some of the Oak pupils have decided that they want to work out who made the biggest improvement on their times tables scores when compared to their starting scores.

Rank them in order from the smallest to the biggest improvement.



2) Three different brands are claiming that they have reduced the sugar in their cereal by at least 30%.

a) Which brand cannot claim this? *Brand C*

b) Which brand has reduced the sugar in their cereal by the most? *Brand A*



$$100 - 9.2 \div 13.3 \times 100 = 30.8\% \text{ (1 d.p.)}$$

$$100 - 8.4 \div 12 \times 100 = 30\%$$

$$100 - 6.6 \div 9.4 \times 100 = 29.8\% \text{ (1 d.p.)}$$

3) Below are some items that were bought and sold by a market stall holder. Rank the items based on the percentage change.



Item	Purchase price	Selling price	Percentage change
Mobile phone	£85	£87.55	3% increase
Sunglasses	£8	£7.84	2% decrease
Tablet	£125	£137.50	10% increase
Shoes	£14	£13.58	3% decrease
Cap	£6	£6.48	8% increase

Order: shoes, sunglasses, mobile phone, cap and tablet (or reverse).

